

Ex. 13. If $x + y = 23$ and $xy = 126$, what is the value of $x^2 + y^2$?

(Bank Recruitment, 2010)

Sol. $(x + y) = 23 \Rightarrow (x + y)^2 = (23)^2 = 529 \Rightarrow x^2 + y^2 + 2xy = 529$
 $\Rightarrow x^2 + y^2 + 2 \times 126 = 529 \Rightarrow x^2 + y^2 = 529 - 252 = 277.$

Ex. 14. If $\frac{a}{b} = \frac{4}{5}$ and $\frac{b}{c} = \frac{15}{16}$, find the value of $\frac{c^2 - a^2}{c^2 + a^2}$.

(M.B.A., 2007)

Sol. $\frac{a}{b} = \frac{4}{5}$ and $\frac{b}{c} = \frac{15}{16} \Rightarrow \frac{a}{b} \times \frac{b}{c} = \frac{4}{5} \times \frac{15}{16} = \frac{3}{4} \Rightarrow \frac{a}{c} = \frac{3}{4} \Rightarrow \frac{c}{a} = \frac{4}{3}.$

$$\therefore \frac{c^2 - a^2}{c^2 + a^2} = \frac{\frac{c^2}{a^2} - 1}{\frac{c^2}{a^2} + 1} = \frac{\left(\frac{c}{a}\right)^2 - 1}{\left(\frac{c}{a}\right)^2 + 1} = \frac{\left(\frac{4}{3}\right)^2 - 1}{\left(\frac{4}{3}\right)^2 + 1} = \frac{\left(\frac{16}{9} - 1\right)}{\left(\frac{16}{9} + 1\right)} = \frac{7}{9} \times \frac{9}{25} = \frac{7}{25}.$$

Ex. 15. If $a = \frac{4xy}{x+y}$, find the value of $\frac{a+2x}{a-2x} + \frac{a+2y}{a-2y}$.

(M.C.A., 2007)

Sol. $\frac{a+2x}{a-2x} + \frac{a+2y}{a-2y} = \frac{\frac{4xy}{x+y} + 2x}{\frac{4xy}{x+y} - 2x} + \frac{\frac{4xy}{x+y} + 2y}{\frac{4xy}{x+y} - 2y} = \frac{4xy + 2x(x+y)}{4xy - 2x(x+y)} + \frac{4xy + 2y(x+y)}{4xy - 2y(x+y)}$
 $= \frac{2x(2y+x+y)}{2x(2y-x-y)} + \frac{2y(2x+x+y)}{2y(2x-x-y)} = \frac{(3y+x)}{(y-x)} + \frac{(3x+y)}{(x-y)}$
 $= \frac{(3x+y)}{(x-y)} - \frac{(3y+x)}{(x-y)} = \frac{(3x+y) - (3y+x)}{(x-y)} = \frac{2x-2y}{x-y} = 2.$

Ex. 16. Find the value of $4 - \frac{5}{1 + \frac{1}{3 + \frac{1}{2 + \frac{1}{4}}}}$.

Sol. Given exp. $= 4 - \frac{5}{1 + \frac{1}{3 + \frac{1}{2 + \frac{1}{4}}}} = 4 - \frac{5}{1 + \frac{1}{3 + \frac{4}{9}}} = 4 - \frac{5}{1 + \frac{1}{(31/9)}}$
 $= 4 - \frac{5}{1 + \frac{9}{31}} = 4 - \frac{5}{(40/31)} = 4 - \frac{5 \times 31}{40} = 4 - \frac{31}{8} = \frac{1}{8}.$

Ex. 17. If $\frac{2x}{1 + \frac{1}{1 + \frac{x}{1-x}}} = 1$, then find the value of x .

Sol. We have: $\frac{2x}{1 + \frac{1}{(1-x)+x}} = 1 \Leftrightarrow \frac{2x}{1 + \frac{1}{[1/(1-x)]}} = 1 \Leftrightarrow \frac{2x}{1 + (1-x)} = 1$
 $\Leftrightarrow 2x = 2 - x \Leftrightarrow 3x = 2 \Leftrightarrow x = \frac{2}{3}.$

Ex. 18. (i) If $4x + 5y = 83$ and $\frac{3x}{2y} = \frac{21}{22}$, then what is the value of $y - x$?

(Bank P.O., 2008)

(ii) If $\frac{x}{4} - \frac{x-3}{6} = 1$, then find the value of x .